

Revisiting DrivAer Aerodynamics

Updated results integrating Noise Factors and AI

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Siemens Digital Industries Software



Agenda

SRS

Noise Factors

PhysicsAI



SRS

Simone Landi, Giorgio Giangaspero

Velocity Averaging

Previously reported:

$$\langle |\vec{u}| \rangle$$

Mean of velocity magnitude

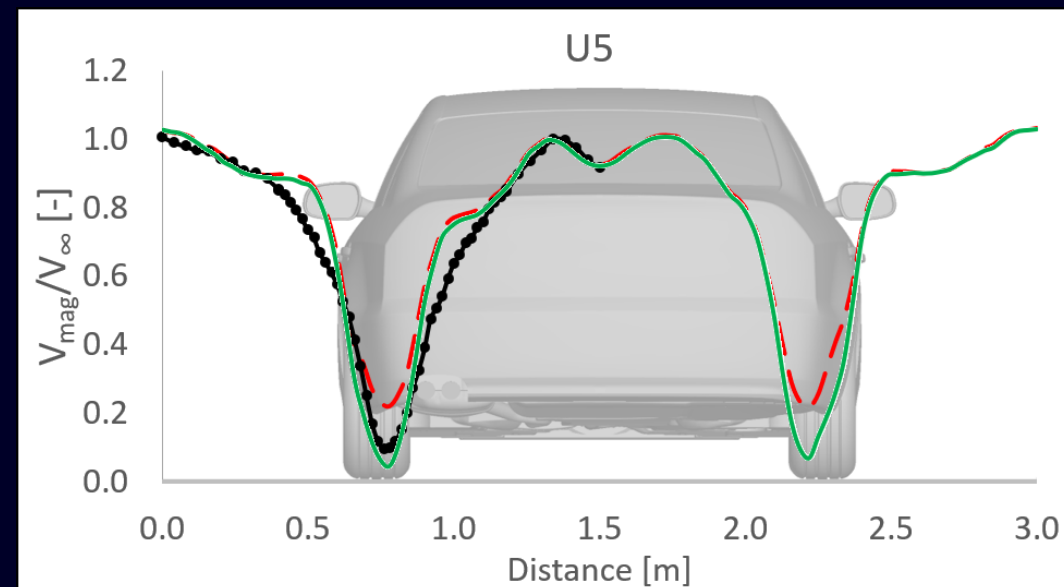
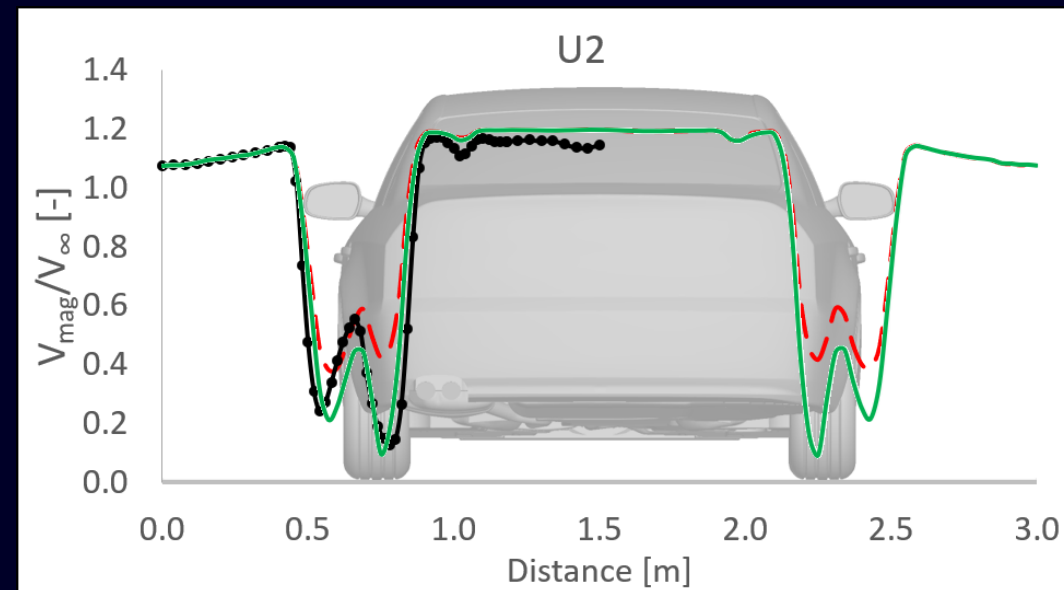
Reporting now:

$$| \langle \vec{u} \rangle |$$

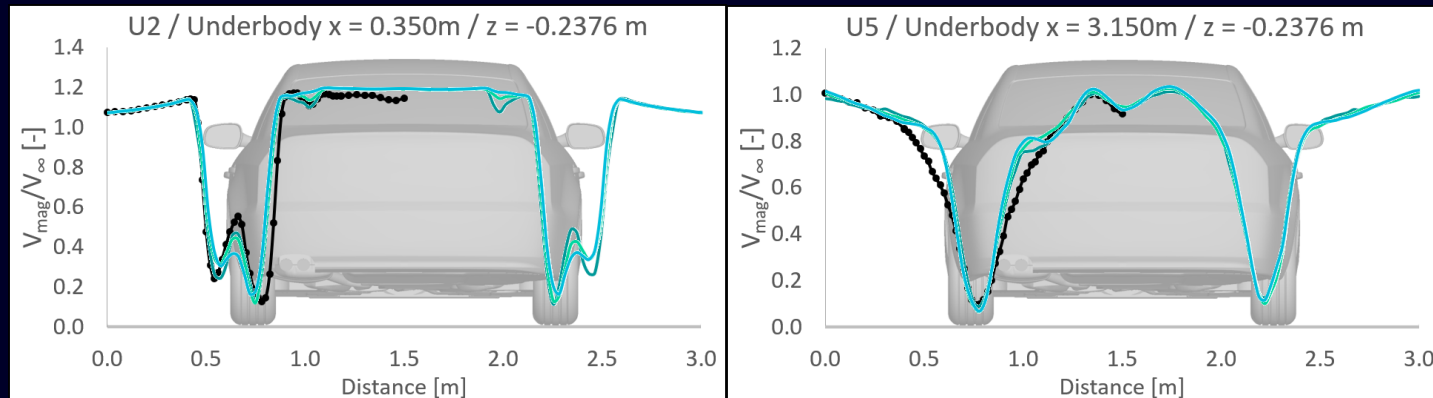
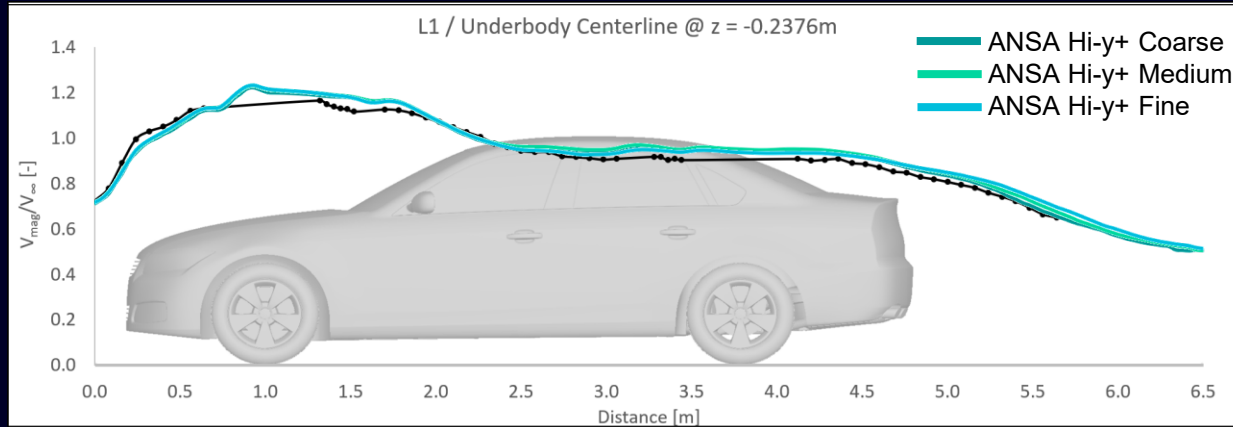
Magnitude of the mean velocity

$$\langle |\vec{u}| \rangle \neq | \langle \vec{u} \rangle |$$

$$\langle |\vec{u}| \rangle \geq | \langle \vec{u} \rangle |$$

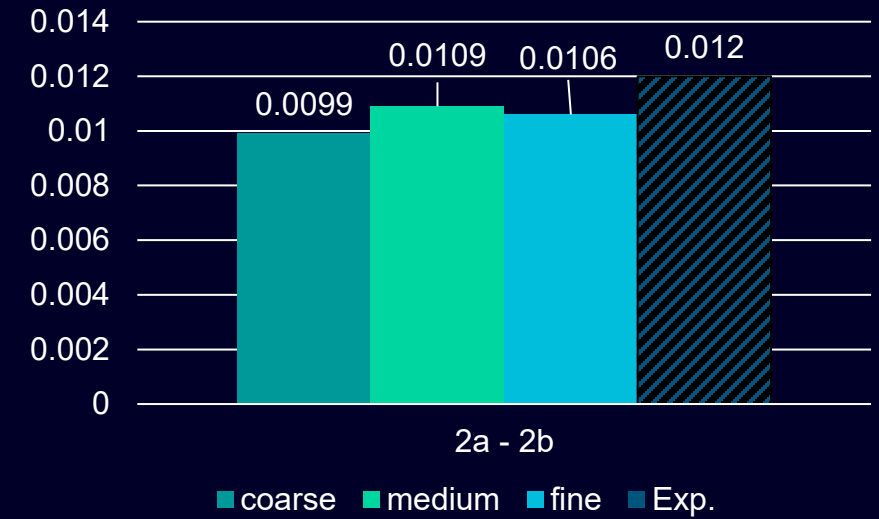


WMLES results, ANSA meshes

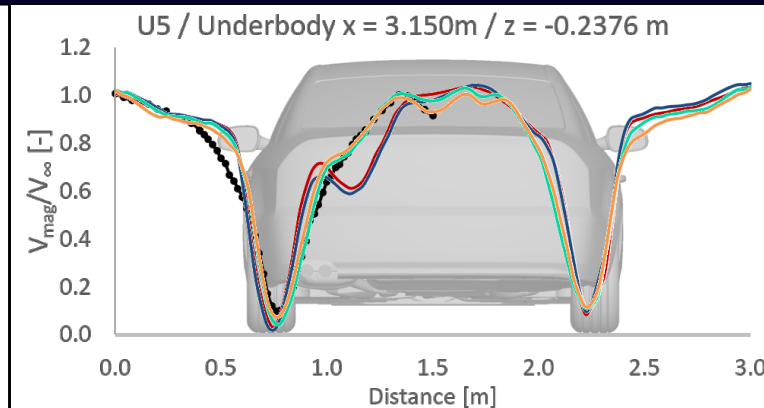
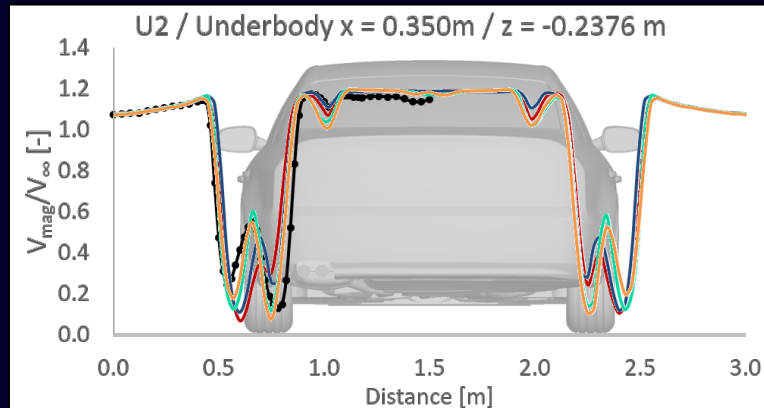
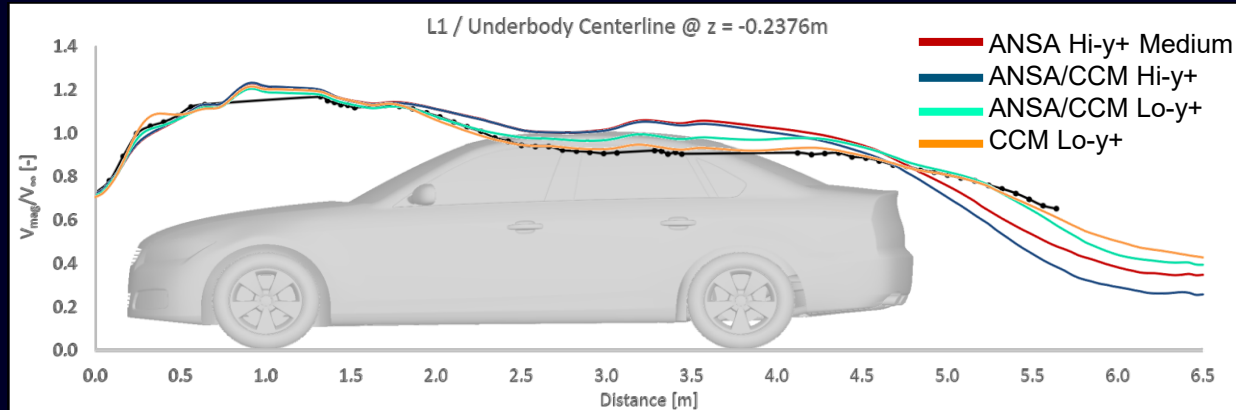


- Time accurate, implicit unsteady
- WALE model
- Total physical time = 4 s
- Averaging time = 3 s
- Transient initialization
- Segregated solver
- MUSCL/CD scheme for convection
- SIMPLEC for pressure/velocity coupling
- $\Delta t = 1\text{e-}4\text{ s}$

Delta Cd

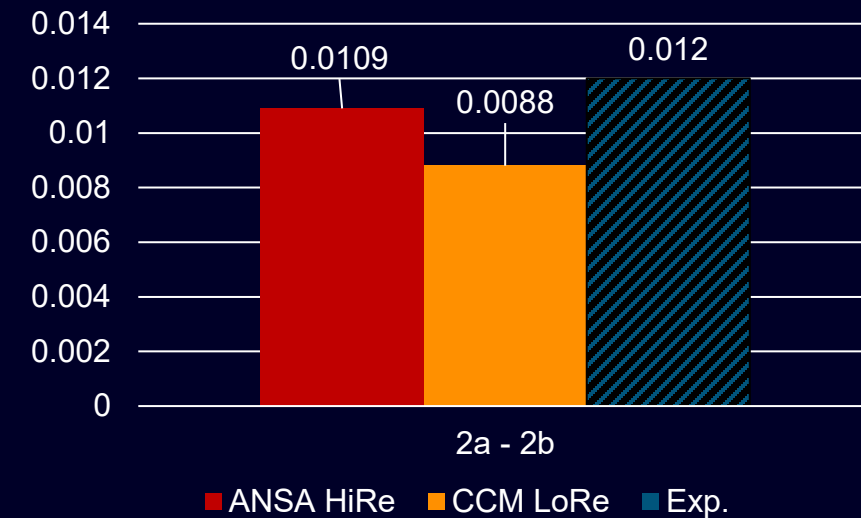


DES and Wall Treatment

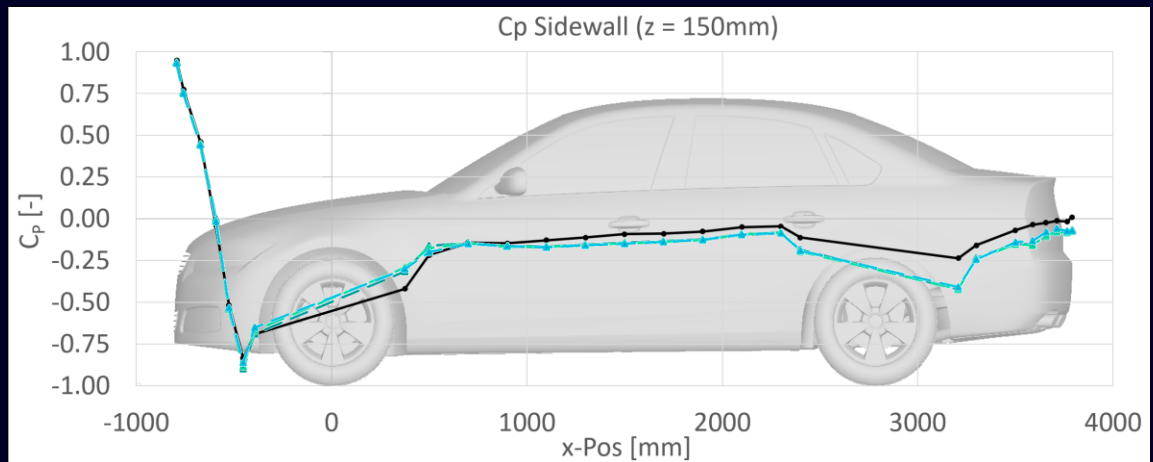
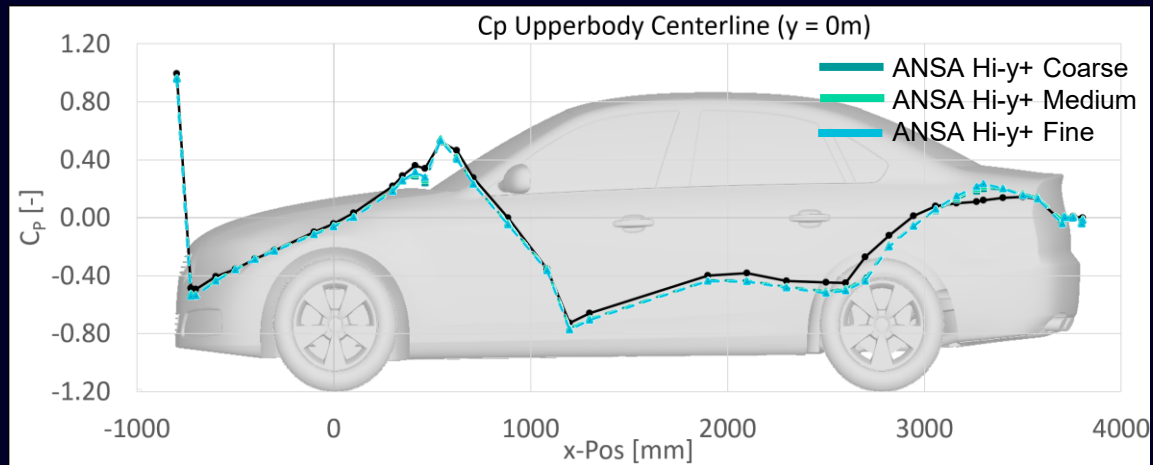


- Time accurate, implicit unsteady
- SST-DDES
- Total physical time = 4 s
- Averaging time = 3 s
- Transient initialization
- Segregated solver
- Hybrid BCD scheme for convection
- SIMPLEC for pressure/velocity coupling
- $\Delta t = 2.5\text{e-}4\text{ s}$

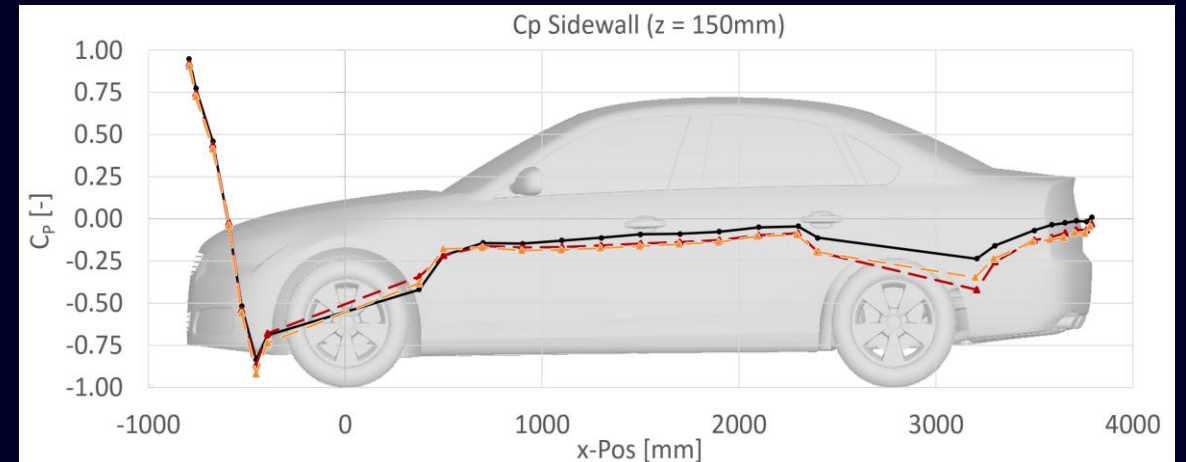
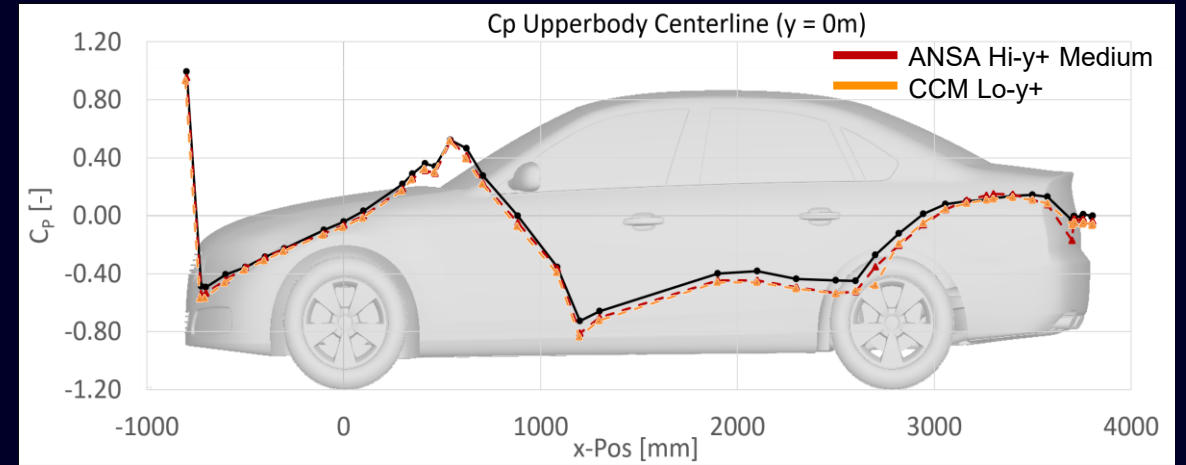
Delta Cd



WMLES results, ANSA meshes

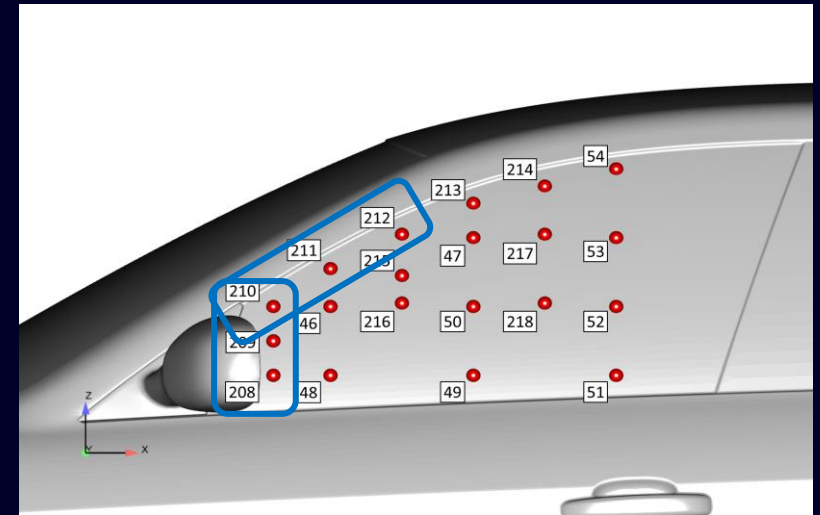
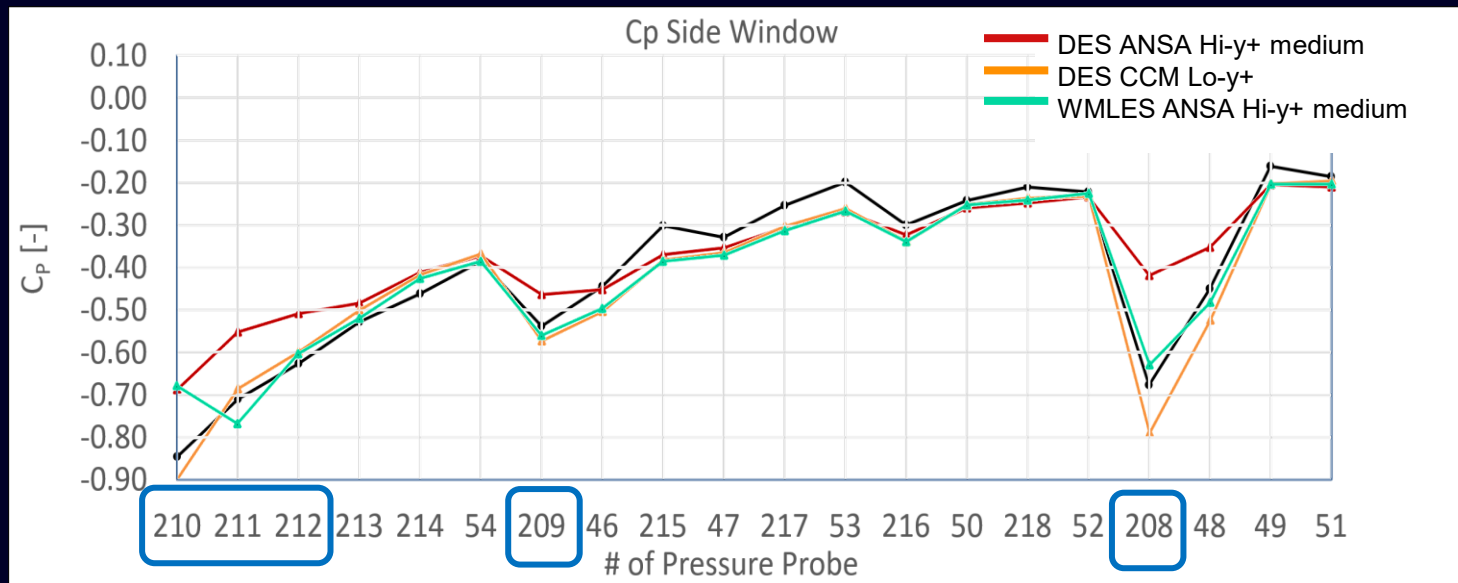


DES results, High/Low Y^+



Side window results

- DES results are significantly improved with Low Y+ mesh, especially in the A-pillar and mirror near wake
 - Probes # 208, 209, 210, 211, 212
- WMLES provides the best result for probe 208, although not doing great on 210 and 211 (A-pillar)





Noise factors

Grzegorz Borowiec

Noise Factors Studies - Can we improve consistency?

Boundary Layer Resolution

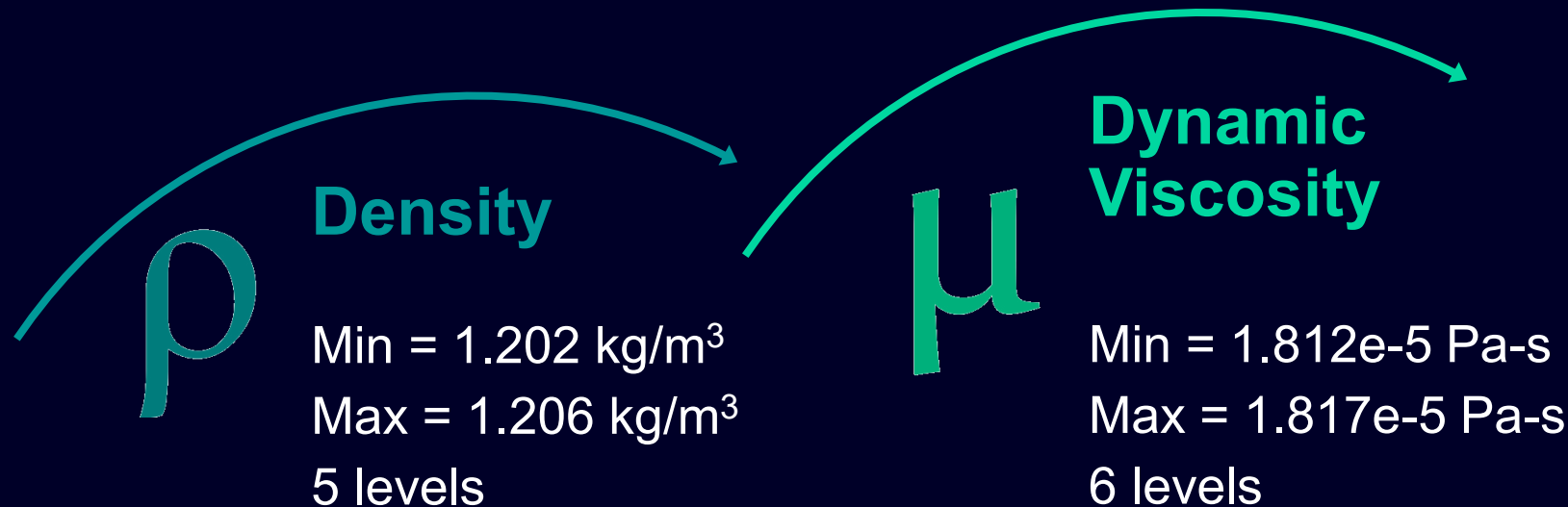
Prism layer settings:

- High Y^+ (7 layers, 0.75 mm nwt, 12 mm thick)
- Low Y^+ (10 layers, 0.03 mm nwt, 10 mm thick)

Deeper convergence per time step

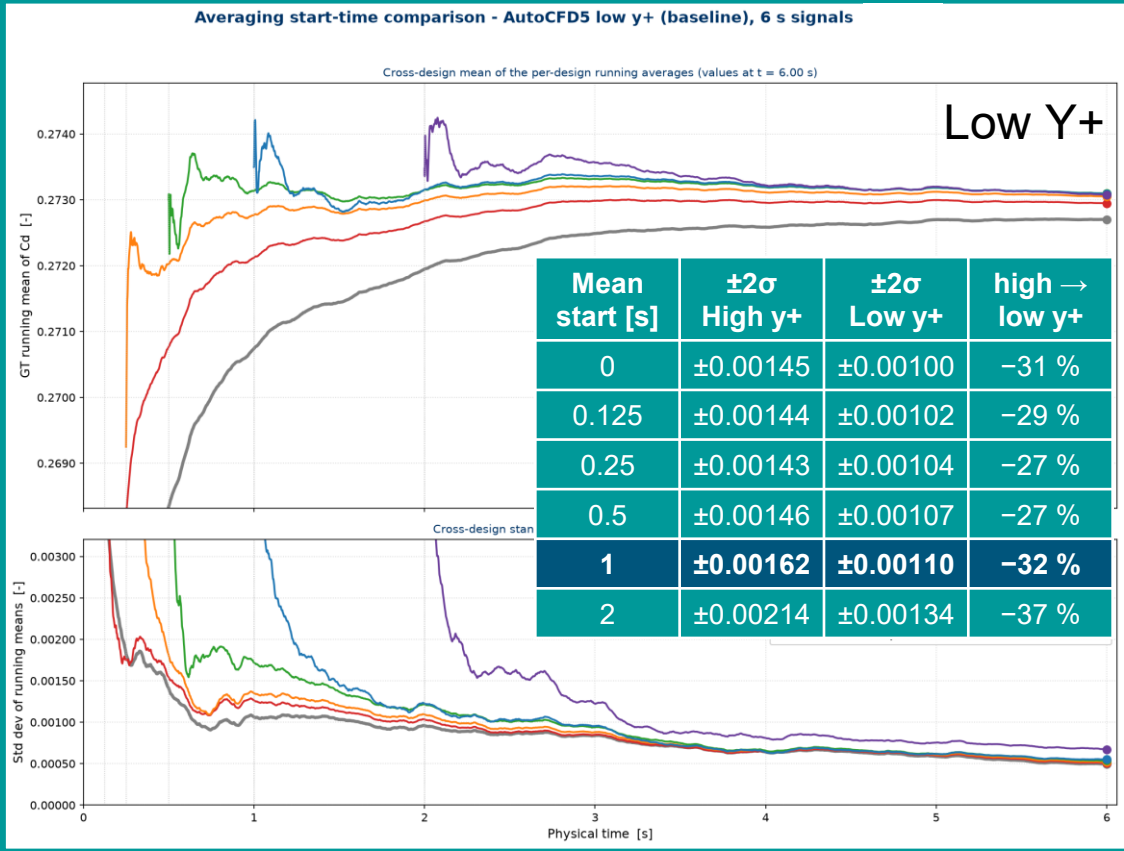
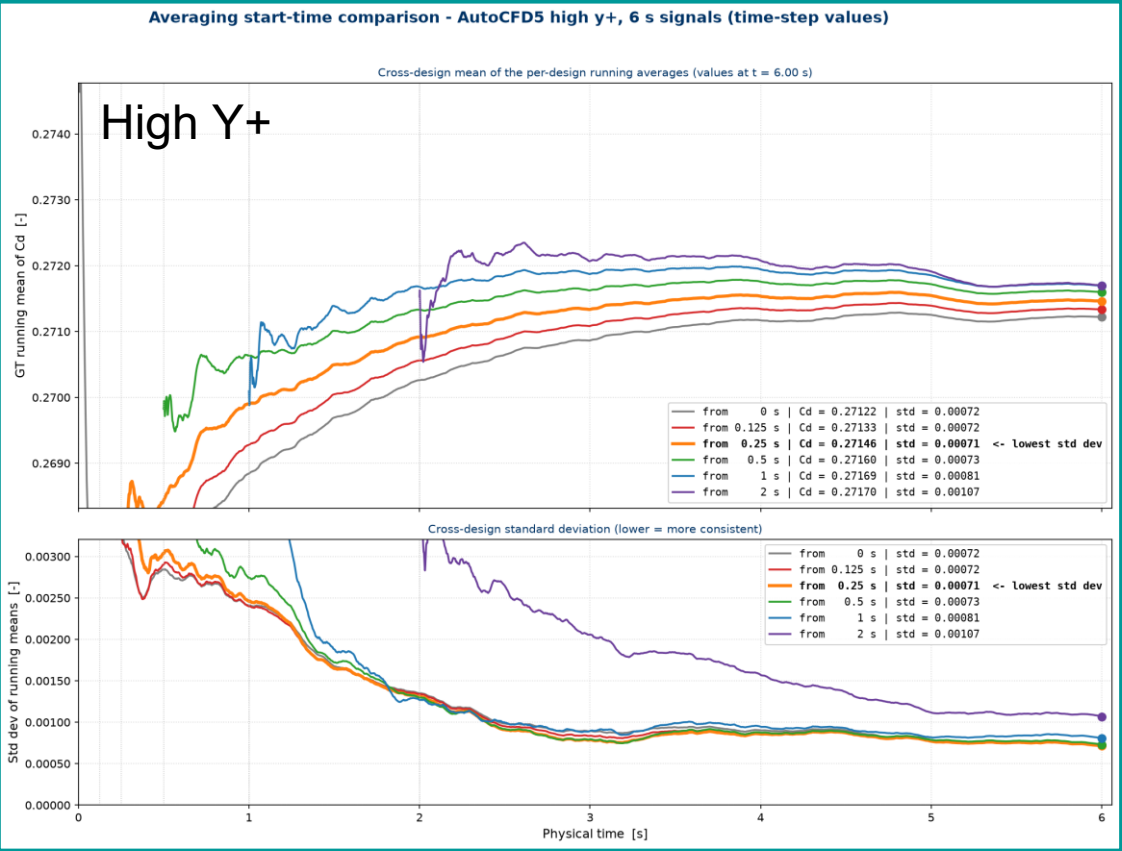
Inner stopping criteria

- Asymptotic $\Delta C_d < 0.25$ cts over 5 samples
- Min 5 ÷ Max 30 iterations per timestep



Noise Factors Studies – Boundary layer resolution

Simcenter STAR-CCM+ 2602 Mixed-precision
Simcenter STAR-CCM+ mesh, incorporating AutoCFD5 refinements
Constant Density, SST-DDES, Transient Initialization
Analyzing mean of time-averaged solutions starting at different times



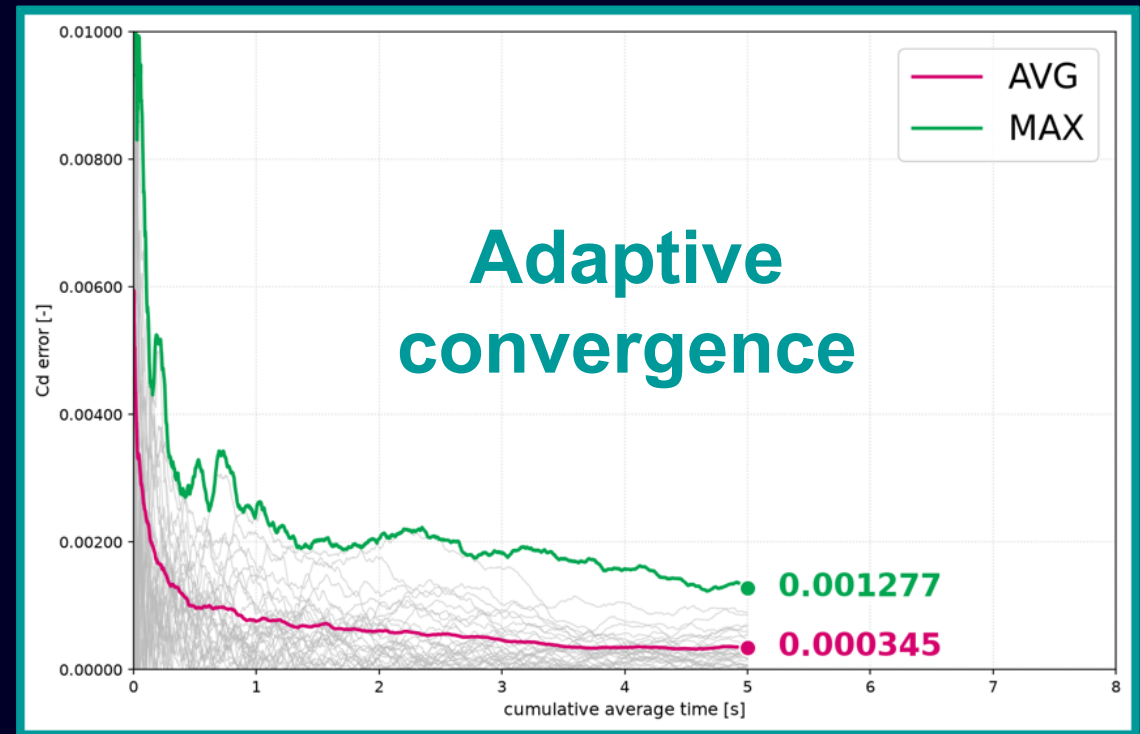
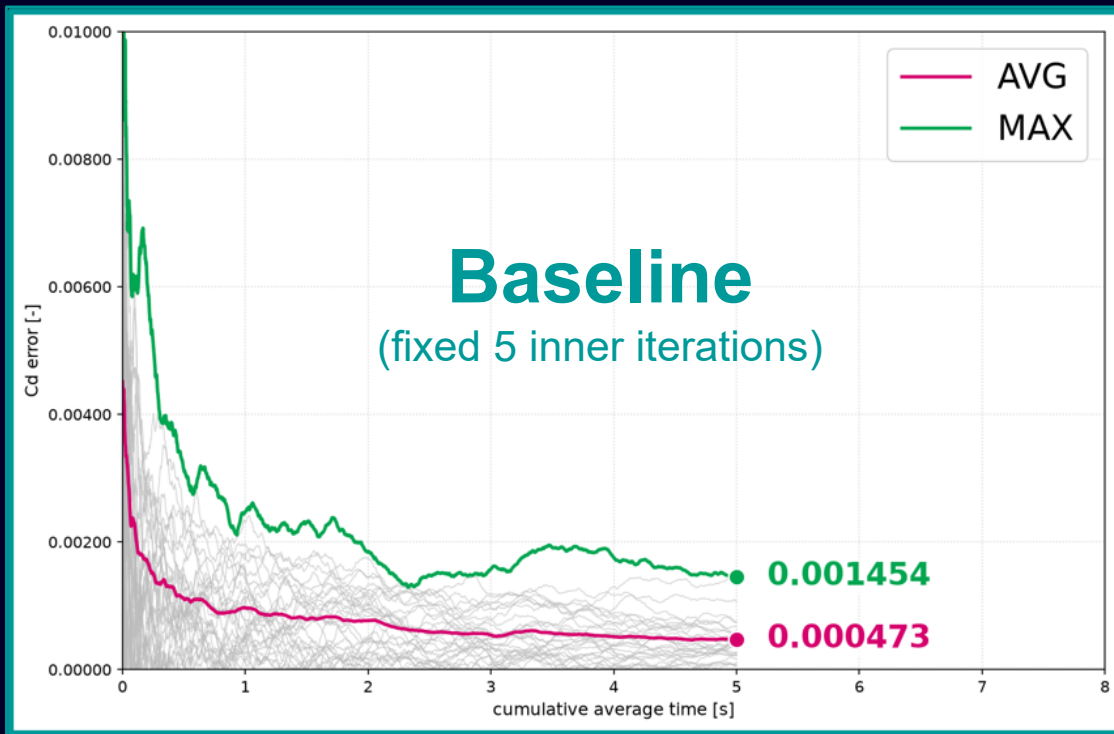
Noise Factors Studies – Adaptive convergence

Simcenter STAR-CCM+ 2602 Mixed-precision

Simcenter STAR-CCM+ mesh, incorporating AutoCFD5 refinements

Constant Density, SST-DDES, Transient Initialization, **Low Y+**

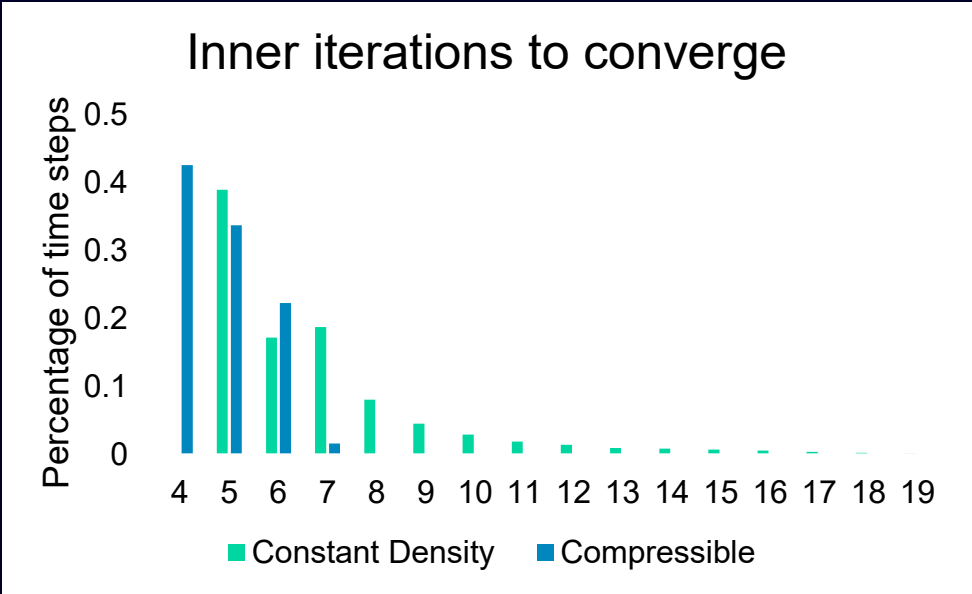
Analyzing average/maximum deviation of all time histories from the overall mean signal



Noise Factors Studies – Adaptive convergence

- Irrespectively of choice of start of averaging time – **adaptive convergence offers better consistency of results**
- Running longer is another effective strategy to minimize the spread, but in our experience adaptive convergence yields not only higher precision but also higher accuracy

Averaging start [s]	$\pm 2\sigma$ baseline	$\pm 2\sigma$ adaptive	Change baseline → adaptive
0	± 0.00106	± 0.00094	-11 %
0.125	± 0.00107	± 0.00096	-10 %
0.25	± 0.00110	± 0.00096	-12 %
0.5	± 0.00113	± 0.00093	-18 %
1	± 0.00119	± 0.00092	-22 %
2	± 0.00143	± 0.00099	-31 %



Constant Density

- 40% within 5 iterations
- 91% within 10 iterations
- 99.8% within 20 iterations

Compressible

- 42% within 4 iterations
- 100% within 8 iterations

+ 59% time*

+ 17% time*

* Compared to 5 fixed inner iterations



PhysicsAI

Toby Emm

DrivAerML – Full Split

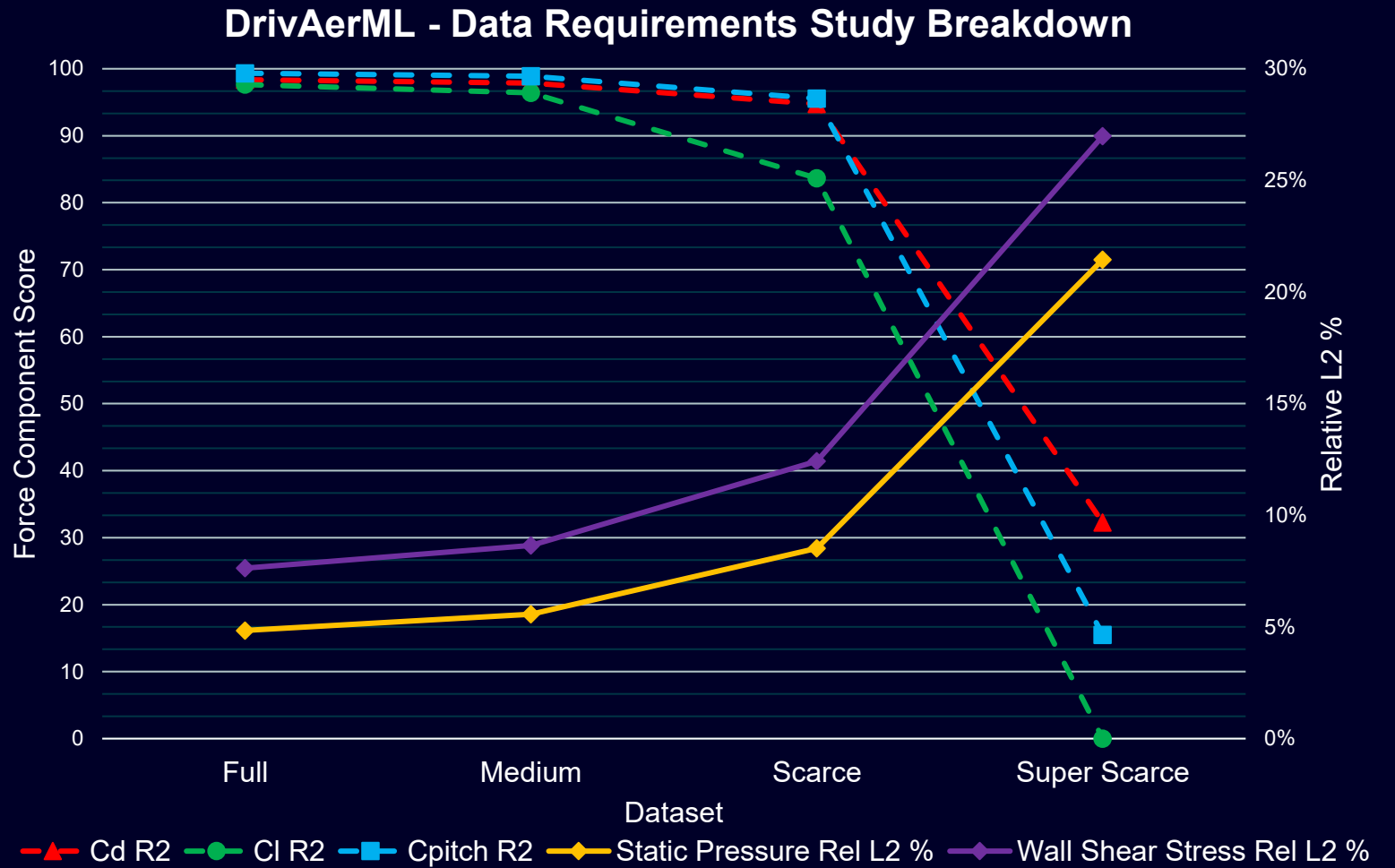
Dataset	Full
Overall Score	50.9
Overall Score (%)	84.9%
Force Score	98.41
Field Score	32.68
Cd R ²	98.4
Cl R ²	97.6
Cpitch R ²	99.3
Static Pressure Rel L2 Score	67.7
Wall Shear Rel L2 Score	61.8
Cp Cut Planes R2	99.7
Static Pressure Rel L2 %	4.8%
Wall Shear Stress Rel L2 %	7.6%
Mean Error (Drag Counts)	2.9
Max Error (Drag Counts)	11.8

Key Features of Model and Training:

- Transformer based architecture
- Uses 2604 public release
- Solution history format input / output
- Mesh sub-sampled at ~0.5%/epoch
- 18h 58m training time (4x A100-40GB)
- 30s per design prediction time (1x L40-46GB)

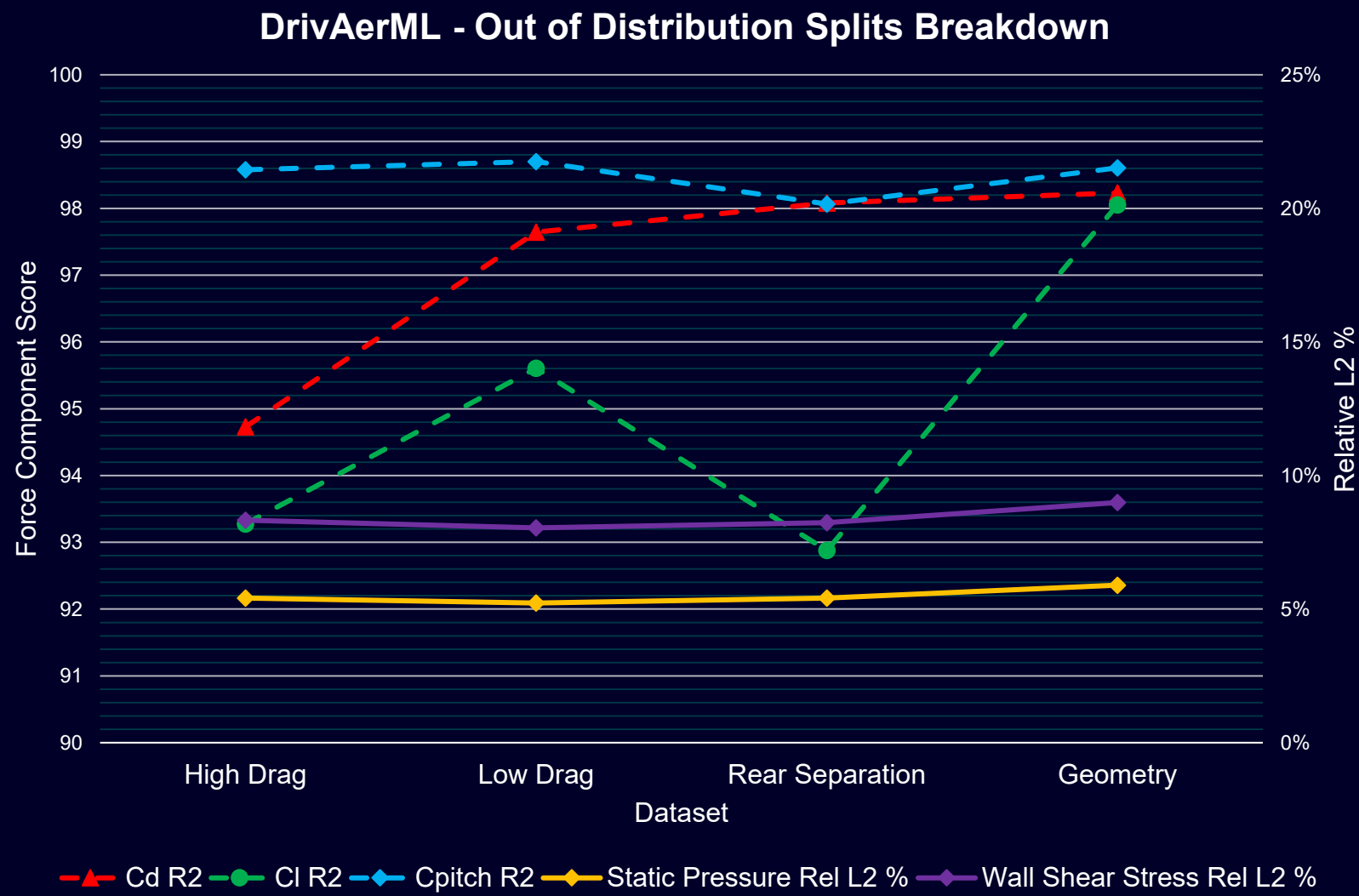
Study 1 – Data Requirements

Dataset	Overall Score	Force Score	Field Score
Full	50.91	98.41	32.68
Medium	49.51	97.78	30.21
Scarce	43.36	92.69	20.53
Super Scarce	15.01	22.45	0.00



Study 2 – Out of Distribution Splits

Dataset	Overall Score	Force Score	Field Score
High Drag	49.20	95.21	30.86
Low Drag	50.08	97.45	31.50
Rear Separation	49.70	97.04	30.94
Geometry	49.14	98.27	29.22



Conclusions

SRS

- Good predictions with WMLES, small mesh dependency
- Low- y^+ meshes confirmed as best practices for DES

Noise Factors

- Lower spread when resolving the boundary layer and when improving convergence within each time-step
- Compressible physics provides faster convergence per time-step

PhysicsAI

- Overall performance is good, particularly force coefficients, relative L2 needs improvement
- Demonstrated ability to generalise well to “out of distribution” splits

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